EXPERIMENT

Softening of hard water

Goal

The goal of this experiment is to understand the concepts of hard and soft water and their implications in daily life. The student will also experiment with the *dissociation of ionic compounds* in water, from which the **hydrogen-donor** nature of the acids and the notion of *pH* are emphasized. Other principles, such as equilibrium and ion exchange, are also presented.
.

Materials

- ☐ 250mL and 50mL beakers
- ☐ cation-exchange resin
- ☐ 6M HCl
- ☐ distilled water (not tab water)
- ☐ Litmus paper
- ☐ 50mL graduated cylinder
- ☐ 125mL Erlenmeyer
- ☐ NH_3/NH_4Cl buffer
- ☐ 0.01M Na_2H_2(EDTA
- ☐ indicator (Eriochrome black mixed with NaCl)
- ☐ dropper solutions (0.10 M $Ca(NO_3)_2$, 0.10 M Na_2CO_3, and 0.10 M $Mg(NO_3)_2$)
- ☐ 0.10 M$NaHCO_3$ solution
- ☐ solid $CaCO_3$

Background

The term **"hard water"** refers to the concentration of certain minerals in the water. This topic concerns **aqueous solutions** in which water is the solvent and minerals are the solutes. These minerals are ionic compounds that dissociate in aqueous solutions producing ions. The hardness of water is an indication of the concentration of Ca^{2+} and Mg^{2+} ions.

Dissolving the insoluble

Ca^{2+} and Mg^{2+} ions originate in limestone ($CaCO_3$, the material of the pyramids) and chalk ($MgCO_3$, used to write on the chalkboard) deposits. However, chalk will not dissolve in water and neither the pyramids dissolve in the rain. How is dissolving these carbonates possible?

The answer is in the air

Water in contact with the air absorbs CO_2 in the atmosphere according to the equation:

$$CO_2\,(g) + H_2O\,(l) \longrightarrow H_2CO_3\,(aq)$$

Where H_2CO_3 is carbonic acid, which reacts with the carbonate to form hydrogen bicarbonate. For the case of $CaCO_3$

$$CaCO_3\,(s) + H_2CO_3\,(aq) \longrightarrow Ca(HCO_3)_2\,(aq)$$

Calcium hydrogen carbonate is also called calcium bicarbonate. Bicarbonates are soluble in water (baking soda is sodium bicarbonate, $NaCO_3$) and, as ionic compounds, they are dissociated into its ions;

$$Ca(HCO_3)_2\,(aq) \longrightarrow Ca^{2+} + 2\,HCO_3{}^-$$

The overall reaction is the 3 previous equations added together:

$$CaCO_3\,(s) + CO_2\,(g) + H_2O\,(l) \longrightarrow Ca^{2+} + 2\,HCO_3{}^-$$

and this is how carbonates dissolve in water.

Sodium hydrogen carbonate can decompose under different conditions. On one hand, in acidic conditions, it produces carbonic acid that quickly decomposes to give carbon dioxide:

$$NaHCO_{3\,(aq)} \xrightarrow{H^+} Na^+ + CO_{2\,(g)} + H_2O_{(l)}$$

On the other side, when heated it produces: $2\,NaHCO_{3\,(aq)} \xrightarrow{\Delta} Na_2CO_3 + CO_{2\,(g)} + H_2O_{(l)}$

Implications in daily life

The presence of these ions in water has several implications for routine actions in our lives. The problem arises when the latter reaction takes place in the opposite direction, releasing CO_2 gas in the process and depositing solid carbonate. This direction is favored when water is heated and explains the white residue in bathrooms and kettles. This is especially a problem for boilers, that suffer from scaling when operating hard water.

The experiment

The experiment consists of four parts: preparing the ion exchange resin, measuring the hardness of a sample of water, reducing the hardness also called softening, of the sample, and testing the hardness of the softened sample. Additionally, a series of simple reactions will be done to test the reactivity of Ca^{2+} and Mg^{2+} ions.

Measuring the hardness of the sample

The concentration of Ca^{2+} ions, or any other metallic cation, is usually measured by titration of the aqueous sample using a sodium salt as the titrant, a solution of known concentration added to the sample. First, an indicator will be added to the sample which will color the solution pink. The indicator chosen is Eriochrome Black T, which will turn blue when the Ca^{2+} ion is removed from the cation-indicator complex:

$$\underset{indicator\ complex\ (pink)}{Ca(In)^{2+}_{(aq)}} \longrightarrow Ca^{2+}_{(aq)} + \underset{indicator\ (blue)}{In_{(aq)}}$$

To achieve the color change H_2EDTA^{2-} will be used. H_2EDTA^{2-} (dihydrogen ethylenediaminetetraacetate) is an anion with a stronger affinity for the Ca^{2+} cations than the indicator. As H_2EDTA^{2-} is added to the solution it will replace the indicator forming a new complex, leaving the indicator alone, which in turn will become blue.

$$H_2EDTA^{2-}_{(aq)} + \underset{(pink)}{Ca(In)^{2+}_{(aq)}} \longleftrightarrow Ca(EDTA)^{2+}_{(aq)} + \underset{(blue)}{In_{(aq)}} + 2\,H^+_{(aq)}$$

Even when the example described above refers to calcium ions, the same chemistry applies to magnesium ions. It is important to note down the exact number of drops required to achieve the endpoint of the titration.

Softening the hard water

After measuring the hardness of the water sample, the next step is to soften the water. *Softening* is the process where ions, particularly calcium and magnesium ions, are removed from the water. When a hard water sample is softened we can refer to it as *softened water*. A cation-exchange resin will be used to soften the water. This resin contains many sulfonic groups, ($-SO_3H$). Every two hydrogens in these acidic groups will be replaced by a metallic cation, Ca^{2+} and Mg^{2+}, in a mechanism called *cation exchange*. Finally, it is necessary to evaluate whether the water sample has been softened. As such the titration performed in Part A will be repeated, but this time with the softened sample. The most relevant parameter is the amount of EDTA added to achieve the endpoint of the titration, to compare this result to the previous titration. This experiment aims to be a qualitative analysis of the water treatment. For that reason, and to simplify the process, the titration will be done using droppers instead of the more precise burette typically used in titrations.

Procedure

Part A. Preparing the resin.

☐ *Step 1:* – Obtain around 5g of the dry, ion exchange resin in a 250 mL beaker. Remember that this resin can be recycled. Do not discard it at the end of the experiment.

☐ *Step 2:* – Add enough 6 M HCl to cover the resin. By adding the acid, the resin is cleaned of metallic cations. To use the ion exchange resin, it must be clean, also known as "in H^+ form".

> ⚠CAUTION!
>
> ⚠ Handle concentrated acid with care to avoid chemical burns.
>
> ⚠ Are you wearing your goggles? Do you know where the eye-washer is?

☐ *Step 3:* – Now you are going to wash the acid away. After 2 minutes, add 200 mL of distilled water to the resin-acid mixture. Allow the resin to settle in the bottom of the beaker.

☐ *Step 4:* – Carefully, pour the liquid off (decant) into another beaker. This liquid is an HCl solution and should be handled and discarded as such.

☐ *Step 5:* – To ensure that all the acid is gone, wash again the resin with 200 mL of distilled water. Allow the resin to settle in the bottom of the beaker and pour the liquid into another beaker.

☐ *Step 6:* – Before you discard the liquid you can test the pH. Dip the tip of a glass stirring rod in the liquid and poke the wet end of the rod into a piece of Blue Litmus paper. If the Litmus paper turns pink, there is still acid, repeat steps 5 and 6. If it does not turn pink, the resin is clean and ready to be used. Reserve the resin for Part B.

Part B. Measure the hardness of a sample of softened water

☐ *Step 1:* – Obtain a little bit more than 150 mL of hard water in a beaker. Hard water is already prepared in the lab.

☐ *Step 2:* – Using a graduated cylinder, add 20.0 mL of hard water to a 125 mL Erlenmeyer. Reserve this water for Part C. Use the rest of the water for this part of the experiment.

☐ *Step 3:* – Transfer the remaining hard water to the beaker with the resin and let it stand for 25 minutes. In the meantime, proceed with Parts C and D.

☐ *Step 4:* – Pour off about 40 mL of the water sample that has been in contact with the resin into a clean beaker. This sample would be referred to as the softened water.

☐ *Step 5:* – Using a graduated cylinder, add 20.0 mL of the softened water to a 125 mL Erlenmeyer.

☐ *Step 6:* – Add 5 mL of an NH_3/NH_4Cl buffer solution (not to be confused with Na_2H_2(EDTA)) to the Erlenmeyer and the tip of a spatula (less than a pea-sized amount) of the indicator (a solid mixture of eriochrome black T and sodium chloride). Adding too many indicators will ruin the experiment. The solution color will be rose pink after the addition of the indicator.

☐ *Step 7:* – Now, we are going to titrate the water sample. Obtain about 20 mL of 0.01 M Na_2H_2(EDTA) (the full name is Ethylenediaminetetraacetic acid) in a 50 mL beaker and a medicine dropper (or a plastic Pasteur pipette). Add the Na_2H_2(EDTA) drop by drop to the mixture while stirring. As the endpoint is approached the solution color will become lavender. Count the drops needed for the analyte to turn blue. When the addition of one drop turns the solution blue, all Ca^{2+} and Mg^{2+} ions have been removed from the sample

☐ *Step 8:* – Clean the resin as described in Part A, and return the resin to the instructor–there is a large beaker ready to collect wet resin.

Part C. Measure the hardness of a water sample

☐ *Step 1:* – While Part B takes place, and the water becomes softer, add 5 mL of a NH_3/NH_4Cl buffer solution (not to be confused with Na_2H_2(EDTA)) to the Erlenmeyer, and the tip of a spatula (less than a pea-sized amount) of the indicator. Rememer that the indicator is just a solid mixture of eriochrome black T and sodium chloride. Adding too many indicators will ruin the experiment. The solution color will be rose pink after the addition of the indicator.

☐ *Step 2:* – Now, we are going to titrate the water sample. Obtain about 20 mL of 0.01 M Na_2H_2(EDTA) in a 50 mL beaker and a medicine dropper (or a plastic Pasteur pipette). Add the Na_2H_2(EDTA) drop by drop to the mixture while stirring. As the endpoint is approached the solution color will become lavender. Count the drops needed for the analyte to turn blue. When the addition of one drop turns the solution blue, all Ca^{2+} and Mg^{2+} ions have been removed from the sample

Part D. Observing reactions with Ca^{2+} and Mg^{2+} ions

☐ *Step 1:* – Obtain 4 test tubes and locate the following solutions: 0.10 M $Ca(NO_3)_2$, 0.10 M Na_2CO_3, 0.10 M$NaHCO_3$, and 0.10 M $Mg(NO_3)_2$.

☐ *Step 2:* – In test tube 1, add 10 drops of $\mathbf{Ca(NO_3)_2}$. Add 10 drops of Na_2CO_3 and write down your observations.

☐ *Step 3:* – In test tube 2, add 10 drops of $\mathbf{Ca(NO_3)_2}$. Add 2 drops of HCl and write down your observations.

☐ *Step 4:* – In test tube 3, add 10 drops of $\mathbf{Ca(NO_3)_2}$. Add 10 drops of 0.1 M $NaHCO_3$ and write down your observations.

☐ *Step 5:* – Now, Heat test tube 3 in the flame of a Bunsen burner. Write down your observations.

☐ *Step 6:* – Now, cool down test tube 3. Add 2 drops of 6M HCl and write down your observations.

☐ *Step 7:* – In test tube 4, add a pea-sized amount of solid $CaCO_3$. Add 2 drops of 6M HCl and write down your observations.

☐ *Step 8:* – Now, repeat the steps above using $\mathbf{Mg(NO_3)_2}$ instead of $\mathbf{Ca(NO_3)_2}$.

☐ *Step 9:* – At this point, complete the remaining steps of Part B.

STUDENT INFO

Name: ____________________ Date: __________

Pre-lab Questions

Softening of hard water

1. A student needs 10 drops of EDTA to reach the equivalency point when titrating a sample of hard water. After the students soften the water with the resin, it requires 12 drops of EDTA to reach the equivalency. Are these results correct? Explain.

2. Compare the sizes used in the experiment: a pea-size and the tip of a spatula. Circle the bigger size.

3. In the experiment, find the chemical name of EDTA.

4. Write the balanced equations for the chemical reactions below. Mind that calcium carbonate is an insoluble compound.
.

$$Na_2CO_{3\,(aq)} \quad + \quad Ca(NO_3)_{2\,(aq)} \quad \longrightarrow$$

$$Ca(NO_3)_{2\,(aq)} \quad \xrightarrow{H^+}$$

$$NaHCO_{3\,(aq)} \quad + \quad Ca(NO_3)_{2\,(aq)} \quad \longrightarrow$$

$$NaHCO_{3\,(aq)} \quad + \quad Ca(NO_3)_{2\,(aq)} \quad \xrightarrow{\Delta}$$

$$NaHCO_{3\,(aq)} \quad + \quad Ca(NO_3)_{2\,(aq)} \quad \xrightarrow{H^+}$$

$$CaCO_{3\,(s)} \quad \xrightarrow{H^+}$$

STUDENT INFO

Name: ______________________ Date: ________

Results
EXPERIMENT

Softening of hard water

Parts B and C: Comparing titrations

Indicate how many drops you used to achieve the endpoint for:

Hard water sample (Part C): ________ drops of Na_2H_2(EDTA)

Softened water sample (Part B): ________ drops of Na_2H_2(EDTA)

Part D: Observing reactions with Ca^{2+} and Mg^{2+} ions

Write down your observations such as gas evolution, the appearance of precipitate, dissolving of precipitate, and no observation.

	$Ca(NO_3)_2$	$Mg(NO_3)_2$
Na_2CO_3		
HCl		
$NaHCO_3$		
$NaHCO_3$+heat		
$NaHCO_3$+heat+HCl		

Solid $CaCO_3$+HCl	

STUDENT INFO

Name: ____________________ Date: __________

Post-lab Questions

Softening of hard water

1. Based on your measurements. Did the ion exchange resin soften the hard water sample? Explain.

2. Write the balanced equations for the chemical reactions below. Mind that magnesium carbonate is an insoluble compound.

$$Na_2CO_{3\,(aq)} \quad + \quad Mg(NO_3)_{2\,(aq)} \quad \longrightarrow$$

$$Mg(NO_3)_{2\,(aq)} \quad \xrightarrow{H^+}$$

$$NaHCO_{3\,(aq)} \quad + \quad Mg(NO_3)_{2\,(aq)} \quad \longrightarrow$$

$$NaHCO_{3\,(aq)} \quad + \quad Mg(NO_3)_{2\,(aq)} \quad \xrightarrow{\Delta}$$

$$NaHCO_{3\,(aq)} \quad + \quad Mg(NO_3)_{2\,(aq)} \quad \xrightarrow{H^+}$$

3. Using the equivalency between drops and mL (1mL=15drops), calculate the number of mL of Na_2H_2(EDTA) needed for the titration of the soft and hard water samples.